

$$f(\mathbf{x}) = \frac{1}{2}x_1^2 + \frac{9}{2}x_2^2.$$

1. The gradient is

$$\nabla f(x_1, x_2) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix} = \begin{pmatrix} x_1 \\ 9x_2 \end{pmatrix}.$$

2. The Hessian is

$$\nabla^2 f(x_1, x_2) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 9 \end{pmatrix}.$$

3. The Hessian is constant, and positive definite. Indeed, as it is a diagonal matrix, its diagonal entries are also its eigenvalues. As they are both positive, the matrix is positive definite. And, consequently, the function f is convex everywhere.

4. The curvature of the function in the direction $\mathbf{d} = (1, 1)^T$ at $\mathbf{x}^+ = (0, 0)^T$ is obtained as

$$\frac{\mathbf{d}^T \nabla^2 f(\mathbf{x}^+) \mathbf{d}}{\mathbf{d}^T \mathbf{d}} = \frac{10}{2} = 5.$$

$$g(\mathbf{x}) = \frac{1}{3}x_1^3 + x_2^3 - x_1 - x_2.$$

1. The gradient is

$$\nabla f(x_1, x_2) = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix} = \begin{pmatrix} x_1^2 - 1 \\ 3x_2^2 - 1 \end{pmatrix}.$$

2. The Hessian is

$$\nabla^2 f(x_1, x_2) = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{pmatrix} = \begin{pmatrix} 2x_1 & 0 \\ 0 & 6x_2 \end{pmatrix}.$$

3. Consider the point $\mathbf{x}_a = (1, 1)^T$. At that point, the Hessian is positive definite and, therefore, the function is convex. If you now consider the point $\mathbf{x}_b = (-1, -1)^T$, the Hessian is negative definite, and the function is concave at $\mathbf{x}_b$. Therefore, the function itself is neither a convex function, nor a concave function.

4. The curvature of the function in the direction $\mathbf{d} = (9, 1)^T$ at the specified point $\mathbf{x}^+ = (9, 1)^T$ is obtained as

$$\frac{\mathbf{d}^T \nabla^2 g(\mathbf{x}^+) \mathbf{d}}{\mathbf{d}^T \mathbf{d}} = \frac{1464}{82} = 17.85.$$